

Faturahman Yudanto

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[Website](#) | [LinkedIn](#) | [GitHub](#) | [Research](#)

RESEARCH INTERESTS

Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Understanding (NLU), and Applied Artificial Intelligence for multidisciplinary domains including healthcare, policy analysis, and engineering.

EDUCATION

Universitas Gadjah Mada

Master of Science in Artificial Intelligence | GPA: 3.96

August 2022 – July 2024

Yogyakarta, Indonesia

- **Thesis:** The Use of Text Normalization on Indonesian User-Generated Content Text Data Augmentation.
- **Research Focus:** Applied Artificial Intelligence, Natural Language Understanding (NLU), Named Entity Recognition (NER), Text Data Augmentation, Aspect-Based Sentiment Analysis, Zero-Shot Text Classification, Text Mining.

Universitas Gadjah Mada

Bachelor of Engineering in Information Engineering | GPA: 3.38

August 2016 – August 2020

Yogyakarta, Indonesia

- **Capstone Project:** Indonesian Sign Language (BISINDO) Recognition using Hierarchical Classification.

RECENT PUBLICATIONS & PRESENTED WORKS

- **Yudanto, F.**, Rafikhansa, R., Bahiuddin, I., Bahari, G., & Winarno, A. (2025), An Empirical Study of Retrieval-Augmented Generation for Corrective Maintenance, presented at 11th International Conference of Asian Society for Precision Engineering and Nanotechnology (ASPEN 2025), New Taipei, Taiwan. [Poster Presentation]
- **Yudanto, F.**, Sari, Y., & Zaki, M. Z. A. C. (2024). Climate-NLI: A Model for Natural Language Inference and Zero-Shot Classification on Climate-Related Text. In S. Dita, J.-B. Kim, A. Borlongan, & N. Oco (Eds.), Proceedings of the 38th Pacific Asia Conference on Language, Information and Computation (pp. 600–608). Tokyo University of Foreign Studies. <https://aclanthology.org/2024.paclic-1.57/> [Poster Presentation]
- **Yudanto, F.**, Fahmi, M., Nazhifah, N., Musdholifah, A., Wardoyo, R., & Afiahayati. (2024). Development of expert system for skin diseases based on named-entity recognition (NER) and fuzzy inference. AIP Conference Proceedings, 3201(1), 050003. <https://doi.org/10.1063/5.0231109> [Oral Presentation]
- Fahmi, M., **Yudanto, F.**, Nazhifah, N., Sari, Y., & Afiahayati. (2023). Deep Learning Approach for Aspect-Based Sentiment Analysis on Indonesian Hospitals Reviews. 2023 Eighth International Conference on Informatics and Computing (ICIC), 1–6. <https://doi.org/10.1109/ICIC60109.2023.10381908>
- Ferdiana, R., Dicka, W. F., & **Yudanto, F.** (2022). MOOD DETECTION BASED ON LAST SONG LISTENED ON SPOTIFY. ASEAN Engineering Journal, 12(3), 123–127. <https://doi.org/10.11113/aej.v12.16834>

RESEARCH & PROFESSIONAL EXPERIENCE

AI Engineer

PT Kedata Indonesia Digital

January 2025 – Present

Yogyakarta, Indonesia

- Researched and fine-tuned BERT-based language models for Natural Language Understanding (NLU) tasks, including text classification, Named Entity Recognition (NER), and information retrieval systems.
- Implemented local Large Language Models (LLMs) to enhance document understanding, data retrieval, and automated text summarization capabilities.

Research Assistant

IEMT Lab, Universitas Gadjah Mada

August 2024 – Present

Yogyakarta, Indonesia

- Investigated and applied Large Language Models (LLMs) to improve document and data comprehension within the mechanical engineering domain.

- Fine-tuned computer vision models and developed hybrid classifiers for an AI-powered road defect detection system, managing end-to-end model serving.
- Designed and deployed scalable REST APIs using Python to serve machine learning models for Remaining Useful Life (RUL) predictions.
- Engineered interactive Streamlit dashboards to visualize complex research datasets and facilitate model output analysis.

Research Assistant
Universitas Gadjah Mada

October 2022 – December 2024
Yogyakarta, Indonesia

- Executed machine learning and Natural Language Processing (NLP) experiments, translating academic research into functional applications.
- Implemented advanced NLP solutions, including topic modeling, zero-shot text classification, and Retrieval-Augmented Generation (RAG) pipelines for collaborative research projects.
- Architected backend APIs to reliably host and serve specialized machine learning models in production environments.
- Developed interactive Streamlit analytic dashboards to streamline complex research data visualization and reporting.

Full Stack Developer
PT Kedata Indonesia Digital

August 2020 – September 2023
Yogyakarta, Indonesia

- Engineered end-to-end NLP pipelines for document summarization and intelligent chatbots, leveraging RAG for context retrieval and LLMs for generation.
- Designed and optimized highly scalable Python-based backend architectures and RESTful API services for diverse data analytics products.
- Researched, trained, and fine-tuned BERT-based models tailored for specific Natural Language Understanding (NLU) objectives.

Software Engineer
Microsoft Innovation Center UGM

July 2018 – July 2020
Yogyakarta, Indonesia

- Implemented Azure Cognitive Services: Text Analytics and ML.NET for mood detection from Spotify last song listened.
- Implemented Azure Cognitive Services: Text Analytics for measure sentiment for product reviews comparison.

SELECTED RESEARCH & APPLIED PROJECTS

Wheel Diameter Analytics API
Tech Stack: Python, FastAPI, Scikit-Learn

November 2025 – Present
Machine Learning Engineer

- Wrapped developed machine learning models for wheel flange wear prediction and remaining useful life estimation into a REST API using FastAPI.
- Develop analytics API for providing insights related to wheel condition and RUL estimation.

Prism: RAG Implementation on Rollingstock Manual Document, Reports, and Digital FMEA
Tech Stack: Python, Streamlit, Ollama, Sentence Transformers

August 2025 – December 2025
AI Engineer

- Developed a Proof-of-Concept for Retrieval-Augmented Generation (RAG) implementation system for rolling stock maintenance documentation
- Created a Streamlit-based chat interface for interactive querying and information retrieval from manual documents, maintenance reports, and digital Failure Mode Effect Analysis (FMEA) items.
- Implement retrieve-rerank mechanism on document retrieval and LLM-based SQL query generation for database retrieval.

Roadster: AI-Powered Road Monitoring System
Tech Stack: Python, Ultralytics YOLO, OpenCV, FastAPI, PostgreSQL

January 2025 – December 2025
AI Engineer

- Fine-tuned YOLO-based computer vision models for road defect detection
- Developed an end-to-end analytics pipeline of road defect detection, including defect detection, defect type estimation, surface distress index (SDI) estimation, and data visualization served through a REST API.
- Prepare Computer Vision Annotation Tools platform for road defect data annotation management

SistemaNextGen-AI: Hospital Information System
Tech Stack: Python, FastAPI, Ollama, Sentence Transformers, PostgreSQL

October 2024 – April 2025
AI Engineer

- Developed AI pipelines for SistemaNextGen, a hospital information system used by Gadjah Mada University Hospital.
- Implemented automatic discharge summary generation module utilizing Llama3.2 model.
- Developed automatic ICD-10 and ICD-9-CM coding from narrative diagnosis and procedure text.

Koneksi Indonesia: Climate Policy Analysis

2024

Tech Stack: Python, BERTopic, Sentence Transformers, Textnets

Research Assistant

- Performed in-depth analysis of Indonesian climate policy news using topic modeling and text network analysis
- Utilized BERTopic and Sentence Transformers to identify topics within the news corpus and leverage Textnets for visualizing the relationships between terms in each topic.

SFC PLN Web App

2024

Tech Stack: Python, Django-ninja, PostgreSQL, Scikit-Learn

Backend Developer

- Designed and implemented the backend for a Service Fuel Consumption (SFC) prediction tool for diesel generators in collaboration with PLN R&D.
- Developed an API services for data management related to diesel generators SFCs and SFC prediction model management.

TECHNICAL SKILLS

NLP: Natural Language Understanding (NLU), Zero-shot Text Classification, Topic Modeling, LLM Application, Retrieval-Augmented Generation (RAG), Named Entity Recognition (NER).

Applied AI: Applied Computer Vision & Image Processing, Machine Learning Model Development, Model Serving Through APIs.

Programming Languages & Frameworks: Python (PyTorch, Hugging Face Transformers, scikit-learn, NLTK, spaCy), FastAPI, Streamlit, Pandas, Plotly, Ultralytics (YOLO), OpenCV, PostgreSQL.

Tools & Platforms: Google Cloud Platform (GCP), Azure Cognitive Services, API Development.

Language Skills: English (TOEFL PBT 597), Indonesian (Native).